



boredofstudies

2012 Bored of Studies Trial Examinations

Mathematics Extension 2

SOLUTIONS

Disclaimer: These solutions may contain small errors. If any are found, please feel free to contact either Carrotsticks or Trebla on www.boredofstudies.org, regarding them.

Thanks: To Trebla, for his many hours spent verifying solutions and suggesting alternate methods.

Multiple Choice

1. A
2. A
3. D
4. C
5. C
6. B
7. B
8. D
9. A
10. A

Brief Explanations

Question 1 Definition of ellipse.

Question 2 Vector must be longer, since we have the reciprocal of a vector that has modulus less than 1. The arguments must also be subtracted, since we are dividing two vectors.

Question 3 Standard t substitution integral.

Question 4 Recall that $\omega^n = 1 \Rightarrow \omega^n - 1 = 0 \Rightarrow (\omega - 1)(\omega^{n-1} + \omega^{n-2} + \dots + \omega + 1) = 0$. So we therefore have $\sum_{k=0}^{n-1} \omega^k = 0$. Also, a further manipulation $\omega^k + \frac{1}{\omega^k} = \frac{\omega^{2k} + 1}{\omega^k}$ along with $\omega^{n+1} = \omega^n \times \omega^1 = \omega$ yields the final result.

Question 5 Consider $P(x) = (x - k)^n Q(x)$, where $k > 0$ and $n \in \mathbb{N}$.

Question 6 Standard resolution of forces. Note that F points upwards since V is less than the optimal speed.

Question 7 Use the inequality $a^2 + b^2 + c^2 \geq ab + bc + ac$ combined with $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ac)$ to acquire the answer. Note that $a + b + c = 1$.

Question 8 (B) and (C) are not even. (A) diverges.

Question 9 The locus is just $xy \leq 1$.

Question 10 Use the Conjugate Root Theorem to factorise, then the discriminant.

Written Response

Question 11 (a)

First divide numerator and denominator by e^x , allowed since $e^x > 0$ for all $x \in \mathbb{R}$.

$$\begin{aligned}\int_0^1 \frac{e^{2x} - 1}{e^{2x} + 1} dx &= \int_0^1 \frac{e^x - e^{-x}}{e^x + e^{-x}} dx \\ &= \ln(e^x + e^{-x}) \Big|_0^1 \\ &= \ln\left(\frac{e + e^{-1}}{1 + 1}\right) \\ &= \ln\left(\frac{e^2 + 1}{2e}\right)\end{aligned}$$

Question 11 (b) (i)

Cross multiply the expressions.

$$\frac{1}{9\cos^2 x - 4\sin^2 x} = \frac{A(x)(3\cos x + 2\sin x) + B(x)(3\cos x - 2\sin x)}{9\cos^2 x - 4\sin^2 x}$$

But recall that $\sin^2 x + \cos^2 x = 1$, so we therefore have, by equating the numerator:

$$\sin^2 x + \cos^2 x = A(x)(3\cos x + 2\sin x) + B(x)(3\cos x - 2\sin x)$$

Group the sines and cosines:

$$\sin^2 x + \cos^2 x = 2\sin x(A(x) - B(x)) + 3\cos x(A(x) + B(x))$$

So therefore, we have:

$$A(x) + B(x) = \frac{1}{3}\cos x \qquad A(x) - B(x) = \frac{1}{2}\sin x$$

Solving simultaneously, we immediately acquire:

$$A(x) = \frac{1}{2}\left(\frac{1}{3}\cos x + \frac{1}{2}\sin x\right) = \frac{1}{12}(2\cos x + 3\sin x)$$

Similarly:

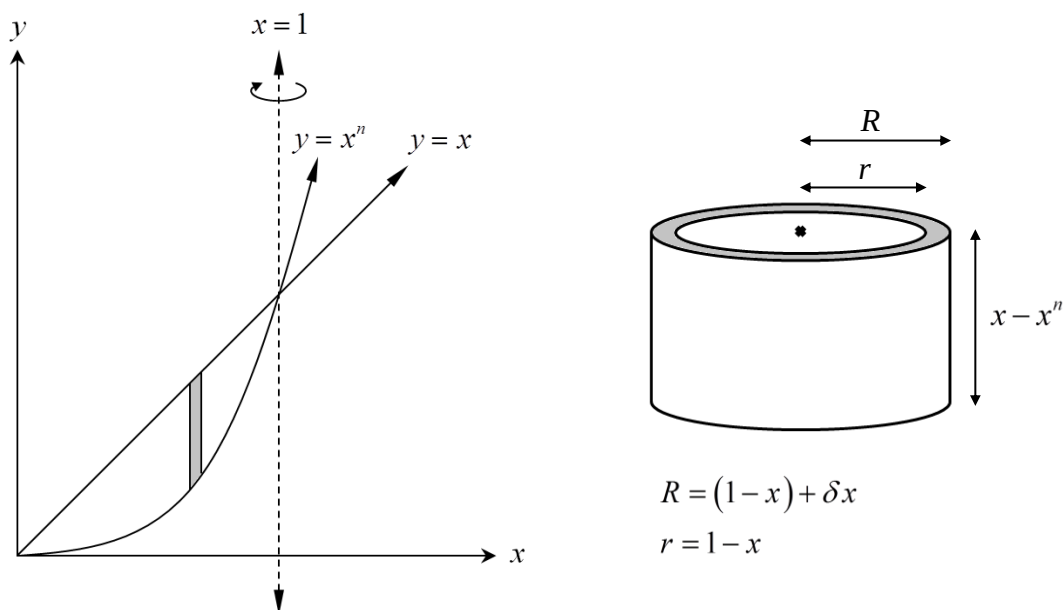
$$B(x) = \frac{1}{2}\left(\frac{1}{3}\cos x - \frac{1}{2}\sin x\right) = \frac{1}{12}(2\cos x - 3\sin x)$$

Question 11 (b) (ii)

Our integral is now:

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \frac{dx}{9\cos^2 x - 4\sin^2 x} &= \frac{1}{12} \int_0^{\frac{\pi}{4}} \left(\frac{2\cos x + 3\sin x}{3\cos x - 2\sin x} + \frac{2\cos x - 3\sin x}{3\cos x + 2\sin x} \right) dx \\&= \frac{1}{12} \int_0^{\frac{\pi}{4}} \left(-\frac{3\sin x - 2\cos x}{3\cos x - 2\sin x} + \frac{-3\sin x + 2\cos x}{3\cos x + 2\sin x} \right) dx \\&= \frac{1}{12} \int_0^{\frac{\pi}{4}} \left(\frac{-3\sin x + 2\cos x}{3\cos x + 2\sin x} - \frac{-3\sin x - 2\cos x}{3\cos x - 2\sin x} \right) dx \\&= \frac{1}{12} \left[\ln(3\cos x + 2\sin x) - \ln(3\cos x - 2\sin x) \right]_0^{\frac{\pi}{4}} \\&= \frac{1}{12} \left[\ln \left(\frac{3\cos x + 2\sin x}{3\cos x - 2\sin x} \right) \right]_0^{\frac{\pi}{4}} \\&= \frac{1}{12} \left[\ln \left(\frac{\frac{3}{\sqrt{2}} + \frac{2}{\sqrt{2}}}{\frac{3}{\sqrt{2}} - \frac{2}{\sqrt{2}}} \right) - \ln(1) \right]_0^{\frac{\pi}{4}} \\&= \frac{1}{12} [\ln(5) - \ln(1)] \\&= \frac{1}{12} \ln 5\end{aligned}$$

Question 11 (c) (i)



Computing the volume of the shell, we have:

$$\begin{aligned}
 \delta V_n &= \pi (R^2 - r^2) (x - x^n) \\
 &= \pi \left(((1-x) + \delta x)^2 - (1-x)^2 \right) (x - x^n) \\
 &= \pi \left((1-x)^2 + 2(1-x)\delta x + (\delta x)^2 - (1-x)^2 \right) (x - x^n) \\
 &= \pi \left(2(1-x)\delta x + (\delta x)^2 \right) (x - x^n) \\
 &\approx 2\pi (1-x) (x - x^n) \delta x \\
 &= 2\pi (x - x^n - x^2 + x^{n+1}) \delta x \\
 &= 2\pi (x - x^2 - x^n + x^{n+1}) \delta x
 \end{aligned}$$

Summing the shells, we have:

$$\begin{aligned}
 V_n &= \lim_{\delta x \rightarrow 0} \sum_{0 \leq x \leq 1} 2\pi (x - x^2 - x^n + x^{n+1}) \delta x &= 2\pi \left[\frac{1}{2} x^2 - \frac{1}{3} x^3 - \frac{1}{n+1} x^{n+1} + \frac{1}{n+2} x^{n+2} \right]_0^1 \\
 &= \int_0^1 2\pi (x - x^2 - x^n + x^{n+1}) \delta x &= 2\pi \left[\frac{1}{2} - \frac{1}{3} - \frac{1}{n+1} + \frac{1}{n+2} \right] \\
 &= 2\pi \int_0^1 (x - x^2 - x^n + x^{n+1}) \delta x &= 2\pi \left[\frac{1}{6} - \frac{1}{n+1} + \frac{1}{n+2} \right]
 \end{aligned}$$

Question 11 (c) (ii)

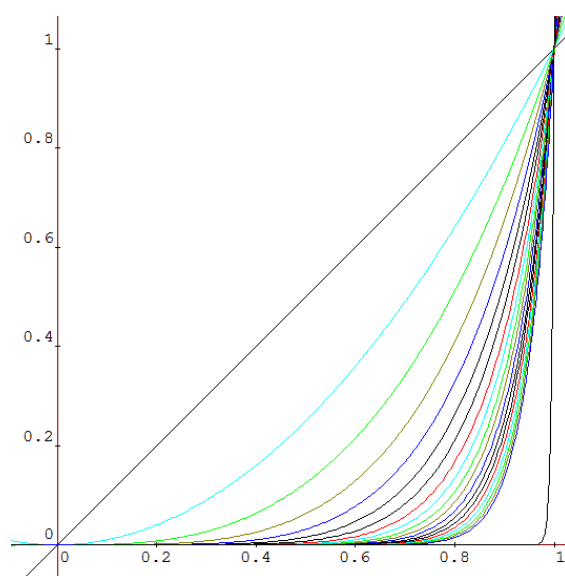
Take the limit as $n \rightarrow \infty$.

$$\begin{aligned}\lim_{n \rightarrow \infty} V_n &= 2\pi \times \frac{1}{6} \\ &= \frac{\pi}{3}\end{aligned}$$

Question 11 (c) (iii)

We observe what happens to $y = x^n$ as $n \rightarrow \infty$.

Since $0 \leq x \leq 1$, we have $0 \leq y \leq 1$. We know that as n gets large, the curve $y = x^n$ gets lower, in the domain $0 \leq x \leq 1$.

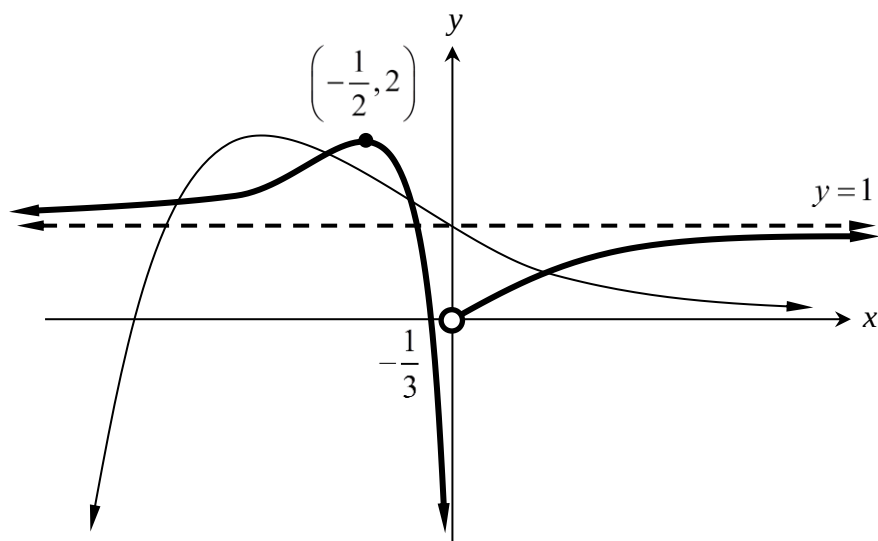


Hence we see that if n gets large, then the area between the two curves will approach a right-angled triangle (the lone curve on the right, in the above diagram, is $y = x^{200}$).

So if we rotate this ‘triangle’ about the line $x = 1$, we will acquire a cone with unit base radius, and unit height.

Using the formula $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$, where $r = 1$ and $h = 1$, we indeed acquire $V_{\text{cone}} = \frac{\pi}{3}$.

Question 11 (d)



Question 12 (a) (i)

Differentiate the expression implicitly.

$$\sqrt{x} + \sqrt{y} = \sqrt{c}$$

$$\frac{1}{2\sqrt{x}} + \frac{y'}{2\sqrt{y}} = 0$$

$$\frac{1}{\sqrt{x}} + \frac{y'}{\sqrt{y}} = 0$$

$$\frac{y'}{\sqrt{y}} = -\frac{1}{\sqrt{x}}$$

$$y' = -\frac{\sqrt{y}}{\sqrt{x}}$$

$$= -\sqrt{\frac{y}{x}} \quad \square$$

Question 12 (a) (ii)

Using the point-gradient formula:

$$y - y_1 = -\sqrt{\frac{y_1}{x_1}}(x - x_1)$$

The line passes through the point $(a, 0)$:

$$\begin{aligned} -y_1 &= -\sqrt{\frac{y_1}{x_1}}(a - x_1) \\ \sqrt{y_1} &= \frac{1}{\sqrt{x_1}}(a - x_1) \\ a - x_1 &= \sqrt{x_1 y_1} \\ a &= \sqrt{x_1 y_1} + x_1 \\ &= \sqrt{x_1}(\sqrt{x_1} + \sqrt{y_1}) \\ &= \sqrt{x_1} \times \sqrt{c} \end{aligned}$$

Similarly for the point $(0, b)$:

$$\begin{aligned} b - y_1 &= -\sqrt{\frac{y_1}{x_1}}(-x_1) \\ &= \sqrt{x_1 y_1} \\ b &= \sqrt{x_1 y_1} + y_1 \\ &= \sqrt{y_1}(\sqrt{x_1} + \sqrt{y_1}) \\ &= \sqrt{y_1} \times \sqrt{c} \end{aligned}$$

So adding them, we have:

$$\begin{aligned} a + b &= \sqrt{x_1} \times \sqrt{c} + \sqrt{y_1} \times \sqrt{c} \\ &= \sqrt{c}(\sqrt{x_1} + \sqrt{y_1}) \\ &= \sqrt{c} \times \sqrt{c} \\ &= c \quad \square \end{aligned}$$

Alternatively

$$y - y_1 = -\sqrt{\frac{y_1}{x_1}}(x - x_1)$$

We know this passes through $(a, 0)$:

$$\begin{aligned} -y_1 &= -\sqrt{\frac{y_1}{x_1}}(a - x_1) \\ \sqrt{y_1} &= \frac{a - x_1}{\sqrt{x_1}} \\ \sqrt{x_1}\sqrt{y_1} &= a - x_1 \end{aligned} \quad (1)$$

Similarly, it passes through $(0, b)$:

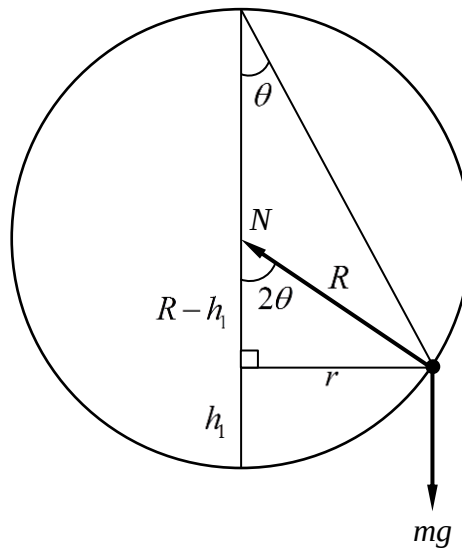
$$\begin{aligned} b - y_1 &= -\sqrt{\frac{y_1}{x_1}}(-x_1) \\ b - y_1 &= \sqrt{x_1}\sqrt{y_1} \end{aligned} \quad (2)$$

Add the two equations:

$$\begin{aligned} 2\sqrt{x_1}\sqrt{y_1} &= a + b - x_1 - y_1 \\ &= a + b - (x_1 + y_1) \\ a + b &= x_1 + 2\sqrt{x_1}\sqrt{y_1} + y_1 \\ &= (\sqrt{x_1} + \sqrt{y_1})^2 \\ &= (\sqrt{c})^2 \\ &= c \quad \square \end{aligned}$$

Question 12 (b)

Consider the following diagram.



Resolving forces vertically: $N \cos 2\theta = mg$

Resolving forces horizontally: $N \sin 2\theta = mr\omega^2$

Divide the two expressions to acquire $\tan 2\theta = \frac{r\omega^2}{g}$. But we also know that $\tan 2\theta = \frac{r}{R-h_1}$, using the right-angled triangle. Equating the two expressions yields:

$$\frac{r\omega^2}{g} = \frac{r}{R-h_1}$$

$$\frac{\omega^2}{g} = \frac{1}{R-h_1}$$

$$\omega^2 = \frac{g}{R-h_1}$$

Our expression is in terms of R . The required expression is in terms of r , so we will use Pythagoras' Theorem to form a relationship between the two variables.

$$\begin{aligned} r^2 &= R^2 - (R-h_1)^2 \\ &= R^2 - R^2 + 2Rh_1 - h_1^2 \\ &= 2Rh_1 - h_1^2 \\ R &= \frac{r^2 + h_1^2}{2h_1} \end{aligned}$$

Substitute this into our expression $\omega^2 = \frac{g}{R-h_1}$:

$$\begin{aligned}\omega^2 &= \frac{g}{\frac{r^2+h_1^2}{2h_1}-h_1} \\ &= \frac{g}{\frac{r^2+h_1^2-2h_1^2}{2h_1}} \\ &= \frac{2gh_1}{r^2-h_1^2}\end{aligned}$$

So therefore to maintain a height greater than h_1 , the particle must have angular velocity

$$\omega \geq \sqrt{\frac{2gh_1}{r^2-h_1^2}}$$

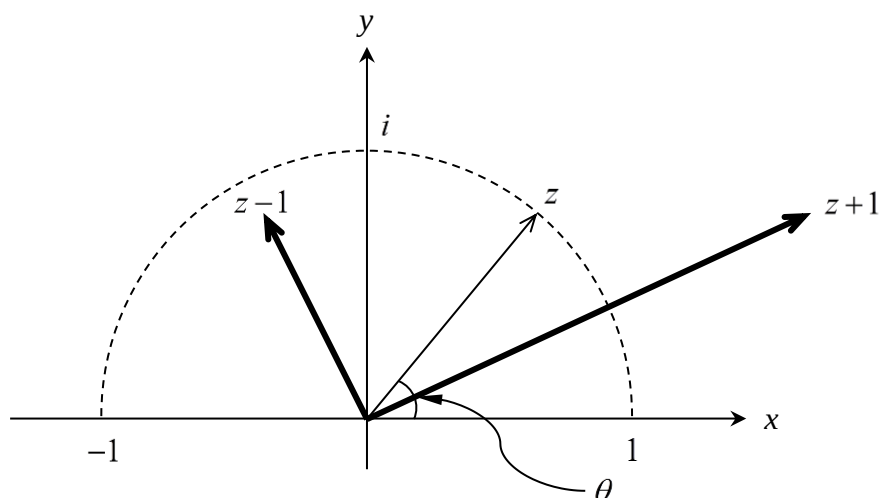
Similarly for h_2 , the particle must have angular velocity $\omega \leq \sqrt{\frac{2gh_2}{r^2-h_2^2}}$.

Therefore we have the required expression:

$$\sqrt{\frac{2gh_1}{r^2-h_1^2}} \leq \omega \leq \sqrt{\frac{2gh_2}{r^2-h_2^2}} \quad \square$$

Question 12 (c) (i)

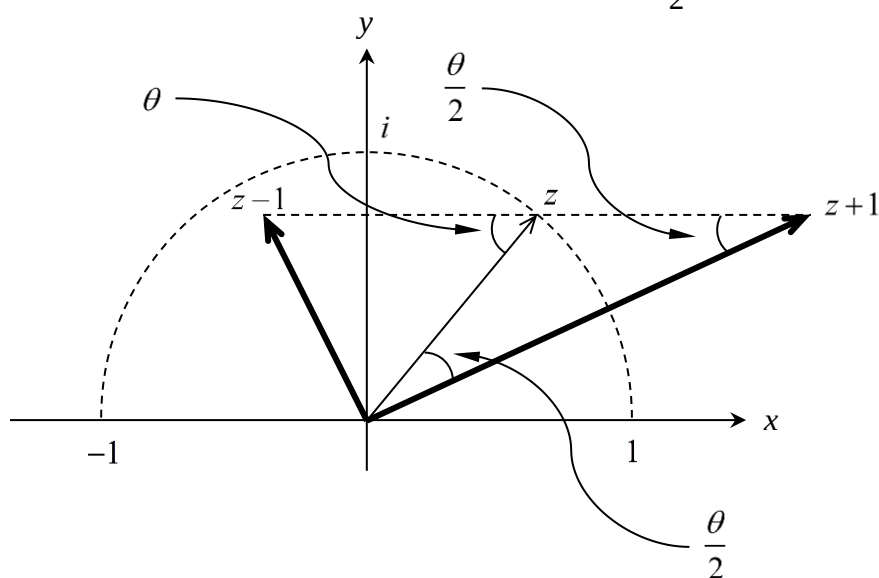
We simply extend the real component one unit forwards and backwards.



Note that as a consequence z , $z-1$ and $z+1$ are now all collinear.

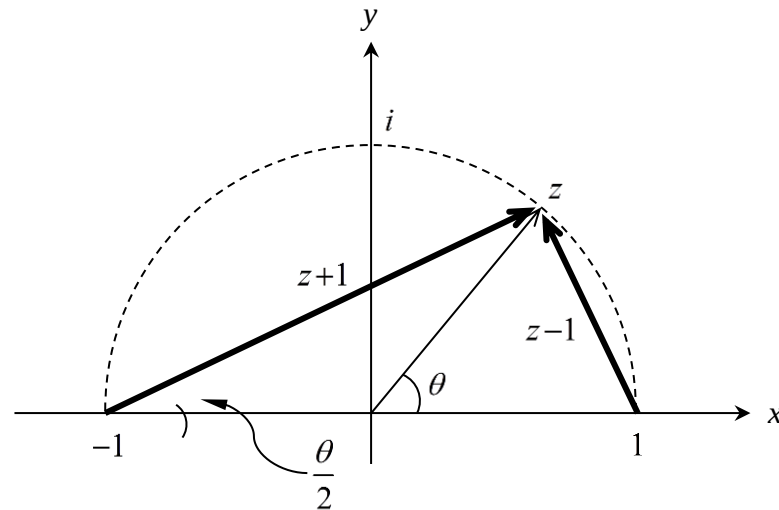
Question 12 (c) (ii)

We will find $\text{Arg}(z-1)$ and $\text{Arg}(z+1)$, and show that they differ by $\frac{\pi}{2}$.



All we need to really spot, is that $\text{Arg}(z) - \text{Arg}(z+1) = \frac{\theta}{2}$, since the triangle enclosed by the origin, z and $z+1$, is an isosceles triangle. A basic angle chase using the exterior angle theorem and the angle sum of triangle quickly yields $\text{Arg}(z-1) - \text{Arg}(z) = \frac{\pi - \theta}{2}$, where it follows immediately that $\text{Arg}(z-1) - \text{Arg}(z+1) = \frac{\pi}{2}$ and thus the result. $\square$

Alternatively



We see immediately that since we have a semi-circle, the angle subtended by the points 1, -1 and z form a right-angled triangle (Thales' Theorem), so therefore we have

$$\text{Arg}(z-1) - \text{Arg}(z+1) = \frac{\pi}{2}, \text{ and thus the result.} \quad \square$$

Question 12 (c) (iii)

The required statement is equivalent to proving that $\left| \frac{z-1}{z+1} \right| = \tan\left(\frac{\theta}{2}\right)$, noting that $\text{Arg}(z) = \theta$.

From diagram, we can easily see that:

$$\begin{aligned} \tan\left(\frac{\theta}{2}\right) &= \frac{|z-1|}{|z+1|} \\ &= \frac{|z-1|}{|z+1|} \quad \square \end{aligned}$$

Question 12 (d) (i)

We know that $|BOC| = \frac{1}{2} \times BC \times OA'$. We also know that $|ABC| = \frac{1}{2} \times BC \times AA'$.

Hence putting the expressions in a fraction, we immediately acquire:

$$\begin{aligned} \frac{|BOC|}{|ABC|} &= \frac{\frac{1}{2} \times BC \times OA'}{\frac{1}{2} \times BC \times AA'} \\ &= \frac{OA'}{AA'} \quad \square \end{aligned}$$

Question 12 (d) (ii)

From (i), we have $\frac{OA'}{AA'} = \frac{|BOC|}{|ABC|}$.

Repeat for other sides to obtain $\frac{OB'}{BB'} = \frac{|AOC|}{|ABC|}$ and $\frac{OC'}{CC'} = \frac{|AOB|}{|ABC|}$.

Add all the expressions to acquire:

$$\begin{aligned} \frac{OA'}{AA'} + \frac{OB'}{BB'} + \frac{OC'}{CC'} &= \frac{|BOC|}{|ABC|} + \frac{|AOC|}{|ABC|} + \frac{|AOB|}{|ABC|} \\ &= \frac{|BOC| + |AOC| + |AOB|}{|ABC|} \\ &= \frac{|ABC|}{|ABC|} \\ &= 1 \quad \square \end{aligned}$$

Question 13 (a) (i)

Consider the line $y = mx + c$. Let this line be tangential to the hyperbola.

To do so, we will substitute the line into the equation of the hyperbola, then let the discriminant be zero

The equation of the hyperbola is $b^2x^2 - a^2y^2 = a^2b^2$. Substitute $y = mx + c$:

$$b^2x^2 - a^2(mx + c)^2 = a^2b^2$$

Expand and re-arrange as a quadratic in terms of x :

$$\begin{aligned} b^2x^2 - a^2(m^2x^2 + 2mcx + c^2) &= a^2b^2 \\ b^2x^2 - a^2m^2x^2 - 2a^2mcx - a^2c^2 &= a^2b^2 \\ x^2(b^2 - a^2m^2) - 2a^2mcx - a^2(b^2 + c^2) &= 0 \end{aligned}$$

Let the discriminant be equal to zero:

$$\begin{aligned} 4a^4m^2c^2 + 4a^2(b^2 - a^2m^2)(b^2 + c^2) &= 0 \\ a^2m^2c^2 + (b^2 - a^2m^2)(b^2 + c^2) &= 0 \\ a^2m^2c^2 + b^4 + b^2c^2 - a^2m^2b^2 - a^2m^2c^2 &= 0 \\ b^4 + b^2c^2 - a^2m^2b^2 &= 0 \\ b^2 + c^2 - a^2m^2 &= 0 \\ c^2 &= a^2m^2 - b^2 \end{aligned}$$

Substitute back into $y = mx + c$:

$$y = mx \pm \sqrt{a^2m^2 - b^2} \quad \square$$

Question 13 (a) (ii)

We will re-arrange $y = mx \pm \sqrt{a^2m^2 - b^2}$ be a quadratic in terms of m .

$$\pm\sqrt{a^2m^2 - b^2} = mx - y$$

$$a^2m^2 - b^2 = (mx - y)^2$$

$$a^2m^2 - b^2 = m^2x^2 - 2mxy + y^2$$

Hence our quadratic is $m^2(a^2 - x^2) + 2mxy - (b^2 + y^2) = 0$.

Since the two tangents are to be perpendicular to each other, we will let the product of roots be equal to -1 , such that $m_1m_2 = -1$.

So immediately, we acquire:

$$\frac{-(b^2 + y^2)}{a^2 - x^2} = -1$$

$$\frac{b^2 + y^2}{a^2 - x^2} = 1$$

$$b^2 + y^2 = a^2 - x^2$$

$$x^2 + y^2 = a^2 - b^2 \quad \square$$

Question 13 (a) (iii)

We know that OT is the radius of the circle, so $OT^2 = a^2 - b^2$.

We also know that $OS = ae$, since S is the focus of the ellipse.

Squaring both sides, we have $OS^2 = a^2e^2$. However, recall the formula $b^2 = a^2(1 - e^2)$:

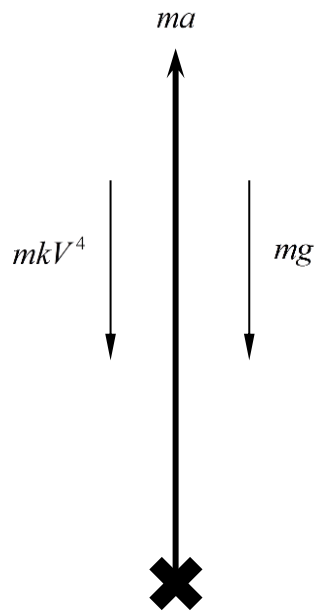
$$\begin{aligned}1 - e^2 &= \frac{b^2}{a^2} \\ e^2 &= 1 - \frac{b^2}{a^2} \\ &= \frac{a^2 - b^2}{a^2}\end{aligned}$$

So therefore:

$$\begin{aligned}OS^2 &= a^2e^2 \\ &= a^2 \left(\frac{a^2 - b^2}{a^2} \right) \\ &= a^2 - b^2 \\ &= OT^2\end{aligned}$$

Thus $OS = OT$, so therefore triangle OTS is isosceles.

Question 13 (b) (i)



Balancing the forces, we have:

$$ma = -mg - mkV^4$$

$$a = -(g + kV^4)$$

$$V \frac{dV}{dx} = -(g + kV^4)$$

Re-arrange and integrate both sides, using appropriate limits:

$$\int_0^{x_t} dx = - \int_U^{V_t} \frac{V}{g + kV^4} dV$$
$$x_t = \int_{V_t}^U \frac{V}{g + kV^4} dV$$

To integrate this, we let $y = V^2$:

$$dy = 2V dV \Rightarrow \frac{dy}{2} = V dV$$

Finding limits:

$$V = U \Rightarrow y = U^2$$

$$V = V_t \Rightarrow y = V_t^2$$

So our integral becomes:

$$\begin{aligned}
x &= \frac{1}{2} \int_{V_t^2}^{U^2} \frac{dy}{g + ky^2} \\
&= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(\frac{y\sqrt{k}}{\sqrt{g}} \right) \Big|_{V_t^2}^{U^2} \\
&= \frac{1}{2\sqrt{kg}} \left[\tan^{-1} \left(\frac{U^2\sqrt{k}}{\sqrt{g}} \right) - \tan^{-1} \left(\frac{V_t^2\sqrt{k}}{\sqrt{g}} \right) \right] \\
&= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(\frac{\frac{U^2\sqrt{k}}{\sqrt{g}} - \frac{V_t^2\sqrt{k}}{\sqrt{g}}}{1 + \frac{U^2\sqrt{k}}{\sqrt{g}} \times \frac{V_t^2\sqrt{k}}{\sqrt{g}}} \right) \\
&= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(\frac{\frac{\sqrt{k}}{\sqrt{g}}(U^2 - V_t^2)}{1 + \frac{kU^2V_t^2}{g}} \right) \\
&= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(\frac{\frac{\sqrt{k}}{\sqrt{g}}(U^2 - V_t^2)}{\frac{g + kU^2V_t^2}{g}} \right) \\
&= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(\frac{\sqrt{k}(U^2 - V_t^2)}{\frac{g + kU^2V_t^2}{\sqrt{g}}} \right) \\
&= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(\frac{\sqrt{kg}(U^2 - V_t^2)}{g + kU^2V_t^2} \right) \quad \square
\end{aligned}$$

Question 13 (b) (iii)

For maximum height, let $V = 0$:

$$\begin{aligned}x &= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(\frac{U^2 \sqrt{kg}}{g} \right) \\&= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(\frac{U^2 \sqrt{k}}{\sqrt{g}} \right) \\&= \frac{1}{2\sqrt{kg}} \tan^{-1} \left(U^2 \sqrt{\frac{k}{g}} \right) \quad \square\end{aligned}$$

Question 13 (b) (iii)

Finding the limit as $U \rightarrow \infty$.

$$\begin{aligned}\lim_{U \rightarrow \infty} \left(\frac{1}{2\sqrt{kg}} \tan^{-1} \left(U^2 \sqrt{\frac{k}{g}} \right) \right) &= \frac{1}{2\sqrt{kg}} \lim_{U \rightarrow \infty} \left(\tan^{-1} \left(U^2 \sqrt{\frac{k}{g}} \right) \right) \\&= \frac{1}{2\sqrt{kg}} \times \frac{\pi}{2} \\&= \frac{\pi}{4\sqrt{kg}}\end{aligned}$$

Question 13 (c) (i)

We will find P_T , then P_S , then arrange them as a ratio.

To find P_T we group the m people and we will have $n-m$ people remaining. However, we must also include the group as a single element, so we have $(n-m+1)!$ permutations for the group + everybody else. We then permute the m people in the group, which yields $m!$ and therefore $P_T = (n-m+1)!m!$.

To find P_S , we will consider the remaining $n-m$ people. If we put these people next to each other in a line, there will be $n-m+1$ ‘gaps’, in which we can place the m people, who are to be separated. This way, we are guaranteed that they are all separated. Hence to choose positions for the m people, we have $\binom{n-m+1}{m}$. We then permute the m people, which gives us $m!$ and then the remaining $n-m$ people, giving us $(n-m)!$ Therefore

$$P_S = \binom{n-m+1}{m} (n-m)!m!$$

So putting it as a ratio, we have:

$$\begin{aligned} \frac{P_S}{P_T} &= \frac{\binom{n-m+1}{m} (n-m)!m!}{(n-m+1)!m!} \\ &= \frac{\binom{n-m+1}{m}}{(n-m+1)} \\ &= \frac{(n-m+1)!}{m!(n-2m+1)!} \\ &= \frac{(n-m+1)!}{(n-m+1)m!(n-2m+1)!} \\ &= \frac{(n-m)!}{m!(n-2m+1)!} \\ &= \frac{1}{m} \times \frac{(n-m)!}{(m-1)!(n-2m+1)!} \\ &= \frac{1}{m} \times \binom{n-m}{m-1} \quad \square \end{aligned}$$

Question 13 (c) (ii)

Suppose that n is even, we can simply substitute $m = \frac{n}{2}$.

$$\begin{aligned}
 \frac{P_s}{P_T} &= \frac{1}{n/2} \times \binom{n-n/2}{n/2-1} \\
 &= \frac{2}{n} \times \binom{n/2}{n/2-1} \\
 &= \frac{2}{n} \times \frac{(n/2)!}{(n/2-1)!(n/2-n/2+1)!} \\
 &= \frac{2}{n} \times \frac{(n/2)!}{(n/2-1)!} \\
 &= \frac{2}{n} \times \frac{n}{2} \\
 &= 1
 \end{aligned}$$

Therefore $P_s = P_T$.

Alternatively

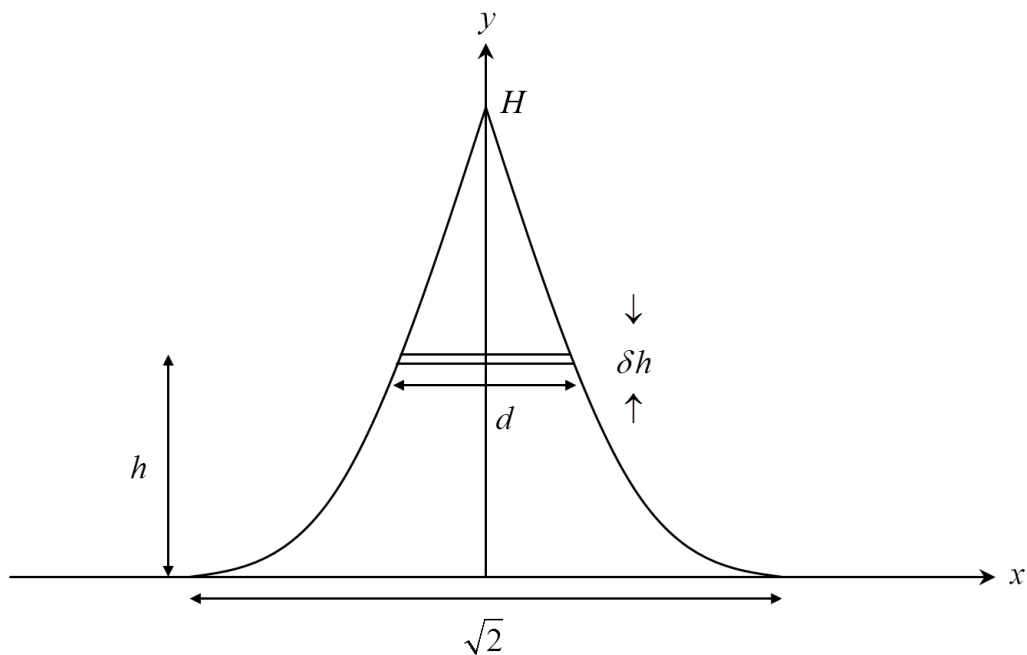
You could substitute $n = 2m$:

$$\begin{aligned}
 \frac{P_s}{P_T} &= \frac{1}{m} \times \binom{2m-m}{m-1} \\
 &= \frac{1}{m} \times \binom{m}{m-1} \\
 &= \frac{1}{m} \times m \\
 &= 1
 \end{aligned}$$

And hence the result. □

Question 14 (a) (i)

If we take the xy plane to be along the diagonal of the base, we have the following situation.



However, recall that the base of the vertex lies on the corner of the base, meaning that the vertex is exactly $\frac{\sqrt{2}}{2}$ units away from the base. We therefore have the equation of the parabolas being:

$$y = \left(x - \frac{\sqrt{2}}{2}\right)^2, \quad 0 \leq x \leq \frac{\sqrt{2}}{2} \quad \quad y = \left(x + \frac{\sqrt{2}}{2}\right)^2, \quad -\frac{\sqrt{2}}{2} \leq x \leq 0$$

To find the value of h in terms of d , we can just use the function. From the diagram, we see that when $x = \frac{d}{2}$, we have $y = h$.

Using the equation of the parabola on the right, we have:

$$\begin{aligned} h &= \left(\frac{d}{2} - \frac{\sqrt{2}}{2}\right)^2 \\ \pm\sqrt{h} &= \frac{1}{2}(d - \sqrt{2}) \quad \dots \text{(Since } d \leq \sqrt{2}, \text{ we take the negative root)} \\ d - \sqrt{2} &= -2\sqrt{h} \\ d &= \sqrt{2} - 2\sqrt{h} \\ &= \sqrt{2}(1 - \sqrt{2h}) \quad \square \end{aligned}$$

Question 14 (a) (ii)

Observe that H is simply the y intercept, so it immediately follows, by letting $x = 0$, that

$$H = \frac{1}{2}.$$

Question 14 (a) (iii)

From the diagonal of the slice, we can deduce the area of the slice.

Since the slice is a square, we can use Pythagoras' Theorem. Let the side length of the square be s .

$$\begin{aligned} s^2 + s^2 &= d^2 \\ 2s^2 &= d^2 \\ &= 2(\sqrt{2h} + 1)^2 \\ s^2 &= (\sqrt{2h} + 1)^2 \end{aligned}$$

So therefore, the volume of the slice is:

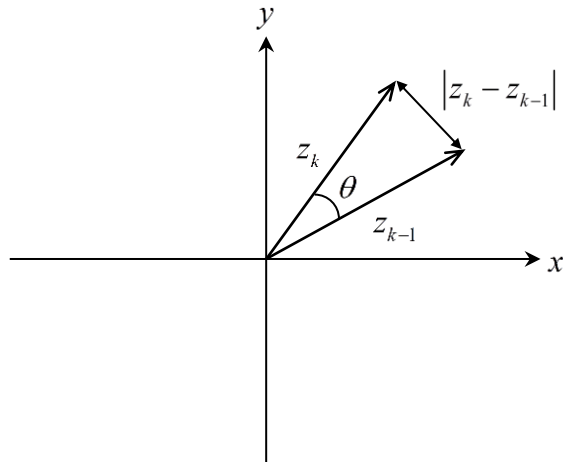
$$\delta V = (\sqrt{2h} + 1)^2 \delta h$$

Integrate from 0 to $\frac{1}{2}$:

$$\begin{aligned} V &= \lim_{\delta h \rightarrow 0} \sum_{0 \leq h \leq 1/2} (\sqrt{2h} + 1)^2 \delta h \\ &= \int_0^{\frac{1}{2}} (\sqrt{2h} + 1)^2 dh \\ &= \int_0^{\frac{1}{2}} \left(2h + 2\sqrt{2}h^{\frac{1}{2}} + 1 \right) dh \\ &= h^2 - 2\sqrt{2} \times \frac{2}{3} h\sqrt{h} + h \Big|_0^{\frac{1}{2}} \\ &= h^2 - \frac{4\sqrt{2}}{3} h\sqrt{h} + h \Big|_0^{\frac{1}{2}} \\ &= \frac{1}{4} - \frac{4\sqrt{2}}{3} \times \frac{1}{2\sqrt{2}} + \frac{1}{2} \\ &= \frac{1}{12} \end{aligned}$$

Question 14 (b) (i)

We will use the Cosine Rule.



Using the Cosine Rule, we have:

$$\begin{aligned}
 |z_k - z_{k-1}|^2 &= |z_k|^2 + |z_{k-1}|^2 - 2|z_k||z_{k-1}|\cos\left(\frac{2\pi}{n}\right) \\
 &= 1 + 1 - 2\cos\left(\frac{2\pi}{n}\right) \quad \dots (\text{since } |z_k| = 1) \\
 &= 2 - 2\cos\left(\frac{2\pi}{n}\right) \\
 |z_k - z_{k-1}| &= \sqrt{2 - 2\cos\left(\frac{2\pi}{n}\right)} \quad \square
 \end{aligned}$$

Alternatively

For simplicity, we let $x = \frac{2k\pi}{n}$ and $y = \frac{2(k-1)\pi}{n}$.

$$\begin{aligned} |z_k - z_{k-1}| &= |\cos x + i \sin x - \cos y - i \sin y| \\ &= |\cos x - \cos y + i[\sin x - \sin y]| \\ &= \sqrt{(\cos x - \cos y)^2 + (\sin x - \sin y)^2} \\ &= \sqrt{\cos^2 x - 2 \cos x \cos y + \cos^2 y + \sin^2 x - 2 \sin x \sin y + \sin^2 y} \\ &= \sqrt{\cos^2 x + \sin^2 x + \cos^2 y + \sin^2 y - 2(\cos x \cos y + \sin x \sin y)} \\ &= \sqrt{2 - 2 \cos \left(\frac{2k\pi}{n} - \frac{2(k-1)\pi}{n} \right)} \\ &= \sqrt{2 - 2 \cos \left(\frac{2k\pi - 2(k-1)\pi}{n} \right)} \\ &= \sqrt{2 - 2 \cos \left(\frac{2k\pi - 2k\pi + 2\pi}{n} \right)} \\ &= \sqrt{2 - 2 \cos \left(\frac{2\pi}{n} \right)} \quad \square \end{aligned}$$

Question 14 (b) (ii)

Geometrically, $|z_k - z_{k-1}|$ is the distance between the endpoints of the two vectors.

So it is the equivalent of the length of one side of the polygon. But we have an n -sided regular polygon, in which case the perimeter would be $P_n = n|z_k - z_{k-1}|$.

Question 14 (b) (iii)

Let $\theta = \frac{2\pi}{n}$, so as $n \rightarrow \infty$, $\theta \rightarrow 0$.

$$\begin{aligned}
 \lim_{n \rightarrow \infty} P_n &= \lim_{n \rightarrow \infty} n \sqrt{2 - 2 \cos \left(\frac{2\pi}{n} \right)} \\
 &= \lim_{\theta \rightarrow 0} \frac{2\pi}{\theta} \sqrt{2 - 2 \cos \theta} \\
 &= \lim_{\theta \rightarrow 0} \frac{2\pi}{\theta} \sqrt{2 - 2 \left(1 - 2 \sin^2 \left(\frac{\theta}{2} \right) \right)} \\
 &= \lim_{\theta \rightarrow 0} \frac{2\pi}{\theta} \sqrt{2 - 2 + 4 \sin^2 \left(\frac{\theta}{2} \right)} \\
 &= \lim_{\theta \rightarrow 0} \frac{2\pi}{\theta} \times 2 \sin \left(\frac{\theta}{2} \right) \\
 &= \lim_{\theta \rightarrow 0} \frac{4\pi}{\theta} \times \sin \left(\frac{\theta}{2} \right) \\
 &= \lim_{\theta \rightarrow 0} 4\pi \times \frac{\sin \left(\frac{\theta}{2} \right)}{\left(\frac{\theta}{2} \right)} \times \frac{1}{2} \\
 &= 4\pi \times \frac{1}{2} \\
 &= 2\pi \quad \square
 \end{aligned}$$

Alternatively

Let $\theta = \frac{\pi}{n}$, so as $n \rightarrow \infty$, $\theta \rightarrow 0$.

$$\begin{aligned}
 \lim_{n \rightarrow \infty} P_n &= \lim_{n \rightarrow \infty} n \sqrt{2 - 2 \cos \left(\frac{2\pi}{n} \right)} \\
 &= \lim_{\theta \rightarrow 0} \frac{\pi}{\theta} \sqrt{2 - 2 \cos 2\theta} \\
 &= \lim_{\theta \rightarrow 0} \frac{\pi}{\theta} \sqrt{2(1 - \cos 2\theta)} \\
 &= \lim_{\theta \rightarrow 0} \frac{\pi}{\theta} \sqrt{4 \sin^2 \theta} \\
 &= \lim_{\theta \rightarrow 0} \frac{2\pi \sin \theta}{\theta} \\
 &= 2\pi \quad \square
 \end{aligned}$$

Question 14 (b) (iv)

The above result is the perimeter of a circle, since the circle is a limiting case of an n -gon when n gets infinitely large.

Question 14 (c) (i)

The point D is a common point.

We know that a circle is unique to a set of three distinct points, in this case D , X and Y . So a circle can indeed be constructed through those points. Since D is a mutual point, CD is also a mutual tangent.

Alternatively

We could do a relatively simple angle chase:

$$\angle CDY = \angle DEB \text{ (Alternate Segment Theorem)}$$

$$\angle DEB = \angle DXY \text{ (Corresponding Angles, since } CW \parallel BE)$$

Hence $\angle CDY = \angle DXY$, satisfying the converse of the Alternate Segment Theorem.

Question 14 (c) (ii)

Using the Secant-Chord Theorem with circle DXY , we have:

$$DC^2 = CY \times CX$$

Similarly for circle DBE , we have:

$$DC^2 = CZ \times CW$$

Equate the two ratios:

$$CY \times CX = CZ \times CW$$

Re-arrange the letters to fit the terms of the question:

$$CY \times XC = ZC \times CW$$

Re-arrange the expressions:

$$\frac{CW}{XC} = \frac{CY}{ZC}$$

But note that $CW = CX + XW$ and $CY = CZ + ZY$. Substitute this in:

$$\begin{aligned}\frac{CX + XW}{XC} &= \frac{CZ + ZY}{ZC} \\ 1 + \frac{XW}{XC} &= 1 + \frac{ZY}{ZC} \\ \frac{XW}{XC} &= \frac{ZY}{ZC} \quad \square\end{aligned}$$

Question 15 (a) (i)

We evaluate $I_{n+1} = \lim_{a \rightarrow \infty} \int_0^a e^{-x} dx$.

$$\begin{aligned}
 I_{n+1} &= \lim_{a \rightarrow \infty} \int_0^a e^{-x} dx. \\
 &= \lim_{a \rightarrow \infty} \left[-e^{-x} \right]_0^a \\
 &= -\lim_{a \rightarrow \infty} \left[e^{-x} \right]_0^a \\
 &= -\lim_{a \rightarrow \infty} \left[e^{-a} - e^0 \right] \\
 &= -(-1) \\
 &= 1 \quad \square
 \end{aligned}$$

Question 15 (a) (ii)

Shift the subscript up by 1 to get $I_{n+1} = \lim_{a \rightarrow \infty} \int_0^a x^n e^{-x} dx$.

Using Integration by Parts, we have:

$$\begin{aligned}
 u &= x^n & dv &= e^{-x} dx \\
 du &= nx^{n-1} dx & v &= -e^{-x} \\
 I_{n+1} &= \lim_{a \rightarrow \infty} -x^n e^{-x} \Big|_0^a + \lim_{a \rightarrow \infty} n \int_0^a x^{n-1} e^{-x} dx \\
 &= \lim_{a \rightarrow \infty} -a^n e^{-a} + nI_n \\
 &= nI_n \quad \square
 \end{aligned}$$

Question 15 (a) (iii)

Substitute into itself recursively:

$$\begin{aligned}
 I_{n+1} &= nI_n \\
 &= n(n-1)I_{n-1} \\
 &= n(n-1)(n-2)I_{n-2} \\
 &= \dots \\
 &= n \times (n-1) \times (n-2) \times \dots \times 2 \times 1 \times I_1
 \end{aligned}$$

But $I_1 = 1$, and hence $I_{n+1} = n!$ $\square$

Question 15 (a) (iv)

Using the previous integral $I_{n+1} = \lim_{a \rightarrow \infty} \int_0^a x^n e^{-x} dx$, we make a substitution.

Let $x = -\ln t$, such that $dx = -\frac{dt}{t}$.

$$x = a \Rightarrow t = e^{-a}$$

$$x = 0 \Rightarrow t = 1$$

But since in the previous integral, we had $a \rightarrow \infty$, we will similarly have $t \rightarrow 0$ and thus the new limit of integration will be some value b getting infinitely small.

$$\begin{aligned} I_{n+1} &= \lim_{b \rightarrow 0} \int_1^b (-\ln t)^n e^{\ln t} \times \frac{-dt}{t} \\ &= \lim_{b \rightarrow 0} \int_1^b (-1)^{n+1} (\ln t)^n t \times \frac{dt}{t} \\ &= \lim_{b \rightarrow 0} \int_1^b (-1)^{n+1} (\ln t)^n dt \\ &= \lim_{b \rightarrow 0} \int_b^1 (-1)^n (\ln t)^n dt \quad \dots (\text{swapping limits uses up a } -1) \\ &= (-1)^n \lim_{b \rightarrow 0} \int_b^1 (\ln t)^n dt \\ &= (-1)^n J_n \end{aligned}$$

So thus we have $(-1)^n J_n = I_{n+1}$. But from (i), we have $I_{n+1} = nI_n$, which implies

$$(-1)^n J_n = nI_n.$$

$$\text{Hence } \frac{I_n}{J_n} = \frac{(-1)^n}{n} \quad \square$$

Alternatively

You could work out the actual reduction formula for J_n , then find the ratio $\frac{I_n}{J_n}$.

$$\begin{aligned}
 u &= (\ln t)^n & dv &= dt \\
 du &= n \times \frac{(\ln t)^{n-1}}{t} dt & v &= t \\
 J_n &= \lim_{b \rightarrow 0} \int_b^1 (\ln t)^n dt \\
 &= \lim_{b \rightarrow 0} \left[t(\ln t)^n \right]_b^1 - n \times \lim_{b \rightarrow 0} \int_b^1 (\ln t)^{n-1} dt \\
 &= \lim_{b \rightarrow 0} \left[0 - b(\ln b)^n \right] - n \times \lim_{b \rightarrow 0} \int_b^1 (\ln t)^{n-1} dt \\
 &= -n \times \lim_{b \rightarrow 0} \int_b^1 (\ln t)^{n-1} dt \quad \dots (\text{see Note}) \\
 &= -n J_{n-1}
 \end{aligned}$$

Computing J_0 :

$$\begin{aligned}
 J_0 &= \lim_{b \rightarrow 0} \int_b^1 (\ln t)^0 dt \\
 &= \lim_{b \rightarrow 0} \int_b^1 dt \\
 &= 1
 \end{aligned}$$

So recursively, we acquire $J_n = (-1)^n n!$.

Placing this as a ratio with I_n :

$$\begin{aligned}
 \frac{I_n}{J_n} &= \frac{(n-1)!}{(-1)^n n!} \\
 &= \frac{(-1)^n}{n} \quad \square \quad \dots \left(\text{since } \frac{1}{(-1)^n} = (-1)^n \right)
 \end{aligned}$$

NOTE

All people, who attempted this question and acquired the result, assumed the limit

$$\lim_{b \rightarrow 0} b(\ln b)^n = 0 \text{ without proof.}$$

However, they were not penalised because the average would have otherwise been too low.

The proof for this, which has been omitted from this document, uses the *Squeeze Law* and *L'Hopital's Principle*.

Question 15 (b) (i)

Summing the areas of the upper rectangles, we have

$$\begin{aligned}
 U &= f(1) \times 1 + f(2) \times 1 + f(3) \times 1 + \dots + f(n-1) \times 1 \\
 &= f(1) + f(2) + f(3) + \dots + f(n-1) \\
 &= \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-1} \\
 &= H_{n-1}
 \end{aligned}$$

Observe that:

$$\begin{aligned}
 E_n &= H_{n-1} - \int_1^n \frac{1}{x} dx \\
 &= H_{n-1} - \ln n
 \end{aligned}$$

Adding $\frac{1}{n}$ to both sides, we acquire:

$$\begin{aligned}
 E_n + \frac{1}{n} &= H_{n-1} - \ln n + \frac{1}{n} \\
 &= H_n - \ln n \quad \square
 \end{aligned}$$

Question 15 (b) (ii)

Taking the limit as $n \rightarrow \infty$, we have:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} (H_n - \ln n) &= \lim_{n \rightarrow \infty} E_n + \frac{1}{n} \\
 &= \gamma
 \end{aligned}$$

Since for large values of n , the term $\frac{1}{n}$ approaches 0.

Question 15 (b) (iii)

We will use Mathematical Induction on n .

Base Case: $n = 1$.

$$\begin{aligned}
 LHS &= S_2 \\
 &= \sum_{r=1}^2 \frac{(-1)^r}{r} \\
 &= -\frac{1}{1} + \frac{1}{2} \\
 &= -\frac{1}{2}
 \end{aligned}
 \qquad
 \begin{aligned}
 RHS &= H_1 - H_2 \\
 &= \sum_{r=1}^1 \frac{1}{r} - \sum_{r=1}^2 \frac{1}{r} \\
 &= \frac{1}{1} - \frac{1}{2} - \frac{1}{1} \\
 &= -\frac{1}{2}
 \end{aligned}$$

Thus the base case has been proven.

Inductive Hypothesis: $n = k$.

$$S_{2k} = H_k - H_{2k}, \text{ for all } k \in \mathbb{N}.$$

Inductive Step: $k \Rightarrow k + 1$.

Required to prove: $S_{2k+2} = H_{k+1} - H_{2k+2}$

$$\begin{aligned}
 S_{2k+2} &= \sum_{r=1}^{2k+2} \frac{(-1)^r}{r} \\
 &= \sum_{r=1}^{2k} \frac{(-1)^r}{r} + \sum_{r=2k+1}^{2k+2} \frac{(-1)^r}{r} \\
 &= S_{2k} - \frac{1}{2k+1} + \frac{1}{2k+2} \\
 &= H_k - H_{2k} - \frac{1}{2k+1} + \frac{1}{2(k+1)} \\
 &= H_k - H_{2k} - \frac{1}{2k+1} + \frac{1}{k+1} - \frac{1}{2(k+1)} \\
 &= \left(H_k + \frac{1}{k+1} \right) - \left(H_{2k} + \frac{1}{2k+1} + \frac{1}{2k+2} \right) \\
 &= H_{k+1} - H_{2k+2} \\
 &= RHS
 \end{aligned}$$

Thus, via proof of induction, true for all $n \in \mathbb{N}$. □

Question 15 (b) (iv)

Recall that $I_{r+1} = r!$

We know that $S_{2n} = H_n - H_{2n}$.

Using previous parts, we have:

$$\lim_{n \rightarrow \infty} (H_n - \ln n) = \gamma \qquad \lim_{n \rightarrow \infty} (H_{2n} - \ln 2n) = \gamma$$

So using the expression $S_{2n} = H_n - H_{2n}$, we have:

$$\begin{aligned} S_{2n} &= (H_n - \ln n) - (H_{2n} - \ln 2n) + (\ln n - \ln 2n) \\ &= (H_n - \ln n) - (H_{2n} - \ln 2n) + \ln \left(\frac{n}{2n} \right) \\ &= (H_n - \ln n) - (H_{2n} - \ln 2n) + \ln \left(\frac{1}{2} \right) \end{aligned}$$

Taking the limit as $n \rightarrow \infty$ of both sides:

$$\begin{aligned} \lim_{n \rightarrow \infty} S_{2n} &= \lim_{n \rightarrow \infty} \left[(H_n - \ln n) - (H_{2n} - \ln 2n) + \ln \left(\frac{1}{2} \right) \right] \\ &= \gamma - \gamma + \ln \left(\frac{1}{2} \right) \\ &= \ln \left(\frac{1}{2} \right) \end{aligned}$$

But we observe that the summands are $\frac{I_r}{J_r}$, so:

$$\begin{aligned} \lim_{n \rightarrow \infty} S_{2n} &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{(-1)^r}{r} \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{I_r}{J_r} \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{(r-1)!}{J_r} \\ &= \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{r!}{J_{r+1}} \end{aligned}$$

$$\text{Hence } \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{r!}{J_{r+1}} = \ln \left(\frac{1}{2} \right) \quad \square$$

Question 16 (a)

We first prove a standard result: $|x - y| \leq |x| + |y|$.

$$\begin{aligned}
 |x - y| &= |x + (-y)| \\
 &\leq |x| + |-y| \\
 &= |x| + |y| \quad \square
 \end{aligned}$$

Using the above inequality, we have

$$\begin{aligned}
 |(z_0 + z_1 + z_2 + \dots + z_n) - (z_1 + z_2 + \dots + z_n)| &\leq |z_0 + z_1 + z_2 + \dots + z_n| + |z_1 + z_2 + \dots + z_n| \\
 |z_0| &\leq |z_0 + z_1 + z_2 + \dots + z_n| + |z_1 + z_2 + \dots + z_n| \\
 &\leq |z_0 + z_1 + z_2 + \dots + z_n| + |z_1| + |z_2| + \dots + |z_n| \\
 |z_0| - |z_1| - |z_2| - \dots - |z_n| &\leq |z_0 + z_1 + z_2 + \dots + z_n| \quad \square
 \end{aligned}$$

Question 16 (b) (i) (1)

$$\begin{aligned}
 |(1-x)R(x)| &= \left| (1-x) \sum_{k=0}^n b_k x^k \right| \\
 &= \left| \sum_{k=0}^n b_k x^k - x \sum_{k=0}^n b_k x^k \right| \\
 &= \left| \sum_{k=0}^n b_k x^k - \sum_{k=0}^n b_k x^{k+1} \right| \\
 &= \left| (b_0 + b_1 x + b_2 x^2 + \dots + b_n x^n) - (b_0 x + b_1 x^2 + b_2 x^3 + \dots + b_n x^{n+1}) \right| \\
 &= \left| b_0 + (b_1 - b_0)x + (b_2 - b_1)x^2 + \dots + (b_n - b_{n-1})x^n - b_n x^{n+1} \right| \\
 &\geq |b_0| - |(b_1 - b_0)x| - |(b_2 - b_1)x^2| - \dots - |(b_n - b_{n-1})x^n| - |b_n x^{n+1}| \\
 &= |b_0| - |b_1 - b_0||x| - |b_2 - b_1||x|^2 - \dots - |b_n - b_{n-1}||x|^n - |b_n||x|^{n+1} \\
 &= |b_0| - (b_0 - b_1)|x| - (b_1 - b_2)|x|^2 - \dots - (b_{n-1} - b_n)|x|^n - |b_n||x|^{n+1} \quad \dots (\text{since } b_n \leq b_{n-1}) \\
 &= b_0 - \sum_{k=1}^n (b_{k-1} - b_k)|x|^k - b_n |x|^{n+1} \quad \square
 \end{aligned}$$

Question 16 (b) (i) (2)

When $|x| < 1$, then the inequality becomes

$$\begin{aligned}
 |(1-x)R(x)| &\geq b_0 - \sum_{k=1}^n (b_{k-1} - b_k) |x|^k - b_n |x|^{n+1} \\
 &> b_0 - \sum_{k=1}^n (b_{k-1} - b_k) - b_n \\
 &= 0 \quad \square \quad \dots (\text{since we have a Telescoping Sum})
 \end{aligned}$$

Question 16 (b) (i) (3)

Since $|(1-x)R(x)| > 0$ when $|x| < 1$, we have $|R(x)| > 0$. This means that since $R(x) \neq 0$ when $|x| < 1$, any root α of $R(x)$ must satisfy $|\alpha| \geq 1$.

Question 16 (b) (ii) (1)

Since β is a root of $S(x)$, we have $S(\beta) = c_0 + c_1\beta + c_2\beta^2 + \dots + c_n\beta^n = 0$.

We now evaluate $T\left(\frac{1}{\beta}\right)$.

$$\begin{aligned}
 T\left(\frac{1}{\beta}\right) &= \sum_{k=0}^n c_{n-k} \frac{1}{\beta^k} \\
 &= \sum_{k=0}^n c_{n-k} \beta^{-k} \\
 &= \frac{1}{\beta^n} \times \sum_{k=0}^n c_{n-k} \beta^{n-k} \\
 &= \frac{1}{\beta^n} \times (c_n \beta^n + c_{n-1} \beta^{n-1} + \dots + c_1 \beta + c_0) \\
 &= 0 \quad \square
 \end{aligned}$$

Question 16 (b) (ii) (2)

Likewise for $R(x)$, we acquire:

$$\left| (1-x)T(x) \right| \geq |c_n| - \sum_{k=1}^n |c_k - c_{k-1}| |x|^{n-k} - |c_0| |x|^{n+1}$$

Note that the exponent is $n-k$ and that we have $c_k - c_{k-1}$ as opposed to $c_{k-1} - c_k$ like before. This is due to the nature of the coefficients (in a sense ‘reversed’) of $T(x)$.

Again, similarly to (i), having $|x| < 1$ yields us another Telescoping Sum, which results in $\left| (1-x)T(x) \right| > 0$. Suppose γ is some root of $T(x)$. We then have $|\gamma| \geq 1$, similarly to (i).

However, recall that $\gamma = \frac{1}{\beta}$ due to (b) (ii) (1), hence $\left| \frac{1}{\beta} \right| \geq 1$ and therefore $|\beta| \leq 1$. $\square$

Question 16 (c) (i)

We know that $A \leq \frac{a_{k-1}}{a_k}$, for all $1 \leq k \leq n$, since A is defined to be the minimum.

Re-arranging, we have:

$$\begin{aligned} \frac{A^k}{A^{k-1}} &\leq \frac{a_{k-1}}{a_k} \\ a_k A^k &\leq a_{k-1} A^{k-1} \end{aligned}$$

And hence we have (noting that all the values are positive since $a_k > 0 \Rightarrow A > 0$):

$$0 < a_n A^n \leq a_{n-1} A^{n-1} \leq \dots \leq a_2 A^2 \leq a_1 A \leq a_0 \quad \square$$

Question 16 (c) (ii)

If we substitute the expression $b_k = a_k A^k$ into the inequality, we have

$$0 < b_n \leq b_{n-1} \leq \dots \leq b_1 \leq b_0$$

Since $P(\omega) = 0$, then

$$\sum_{k=0}^n a_k \omega^k = 0$$

$$\sum_{k=0}^n b_k \frac{\omega^k}{A^k} = 0$$

$$\sum_{k=0}^n b_k \left(\frac{\omega}{A} \right)^k = 0$$

$$R\left(\frac{\omega}{A} \right) = 0$$

The claim, that the polynomial R appears here, is valid because of the inequality with the coefficients.

Hence $\frac{\omega}{A}$ is a root of $R(x)$. $\square$

Question 16 (c) (iii)

Similarly to (c) (ii), we have $a_{k-1} B^{k-1} \geq a_k B^k$ and hence $0 < a_0 \leq a_1 B \leq \dots \leq a_{n-1} B^{n-1} \leq a_n B^n$.

Using the substitution $c_k = a_k B^k$, we have $0 < c_0 \leq c_1 \leq \dots \leq c_{n-1} \leq c_n$. Hence

$$\sum_{k=0}^n a_k \omega^k = 0$$

$$\sum_{k=0}^n c_k \frac{\omega^k}{B^k} = 0$$

$$\sum_{k=0}^n c_k \left(\frac{\omega}{B} \right)^k = 0$$

$$S\left(\frac{\omega}{B} \right) = 0$$

The claim, that we have the polynomial S , is valid here because of the inequality with the coefficients.

And hence $\frac{\omega}{B}$ is a root of $S(x)$.

Question 16 (d)

Since $\beta = \frac{\omega}{B}$ and $|\beta| \leq 1$, we have $\left| \frac{\omega}{B} \right| \leq 1$ and hence $|\omega| \leq B$.

Similarly, since $\alpha = \frac{\omega}{A}$ and $|\alpha| \geq 1$, we have $\left| \frac{\omega}{A} \right| \geq 1$ and hence $|\omega| \geq A$.

Therefore, we obtain the inequality $A \leq |\omega| \leq B$. $\square$